

## Image Formats

---

You would have seen movie files with different extensions - .avi, .mpg, .rm, etc.; document files in .doc, .rtf, or .swx. These are different types of files, meaning data is stored in the file in different ways. Similarly with images we have several formats with different kinds of optimisation and features for different purposes. It is difficult to say just which format is the absolute best - but what we can do is list out the advantages and disadvantages of some common formats so that you know what to use when.

All formats can be classified under two headings, lossless and lossy.

While lossless formats store all of the image information and are great for photo editing work, their downside is the huge file sizes that make it impractical for web applications or for embedding images inside documents or presentations. TIFF is an example of a lossless format, as is Photoshop's native PSD.

Lossy formats compromise some detail from the images in order to reduce file sizes to more manageable levels. The amount of detail that is removed can be set in the image editing software. Sacrificing detail will not make any apparent change to the image at moderate compression levels, but if you are going to work further on the image, the loss of detail will affect the final image quality. A commonly used lossy format is the JPEG format.

### 9.1 The JPEG format

---

The most common format that you would have come across is the JPEG format, often written as ~JPG~. You will find some files with the extension .jpeg, some with .jpg and some others with .jpe - these are all the same. The shortening of the four letter extension to three is in keeping with the DOS limitation of three characters for an extension. JPEG is an acronym for Joint